

INTRODUCTION

- Stellarators do not guarantee particle confinement.
- Existing metrics for the confinement are limited to certain cases
- **Particle tracing using guiding center equations is more general.**

Unfortunately, the computational cost of the integration for many particles limited the use of direct trajectory based optimization historically. Here, **we present the recent work in DESC to reduce the computational burden by the use of GPU via JAX and DIFFRAX libraries.** GPU's are perfect for **embarrassingly parallel jobs**, and hence can integrate thousands of particles simultaneously thanks to their thousands of cores.

We built the infrastructure for particle tracing in DESC and implemented the vacuum guiding center equations. The code design is extremely flexible, and can be extended to different particle trajectory models by just coding the equation without needing to think about the integration.

Particle Tracing API

FIELD OBJECT

- Equilibrium (flux coordinates)
- External Field (lab coordinates)

TRAJECTORY MODEL

- Vacuum Guiding Center
- Finite Beta Guiding Center
- Guiding Center with Viscous Drag
- ...

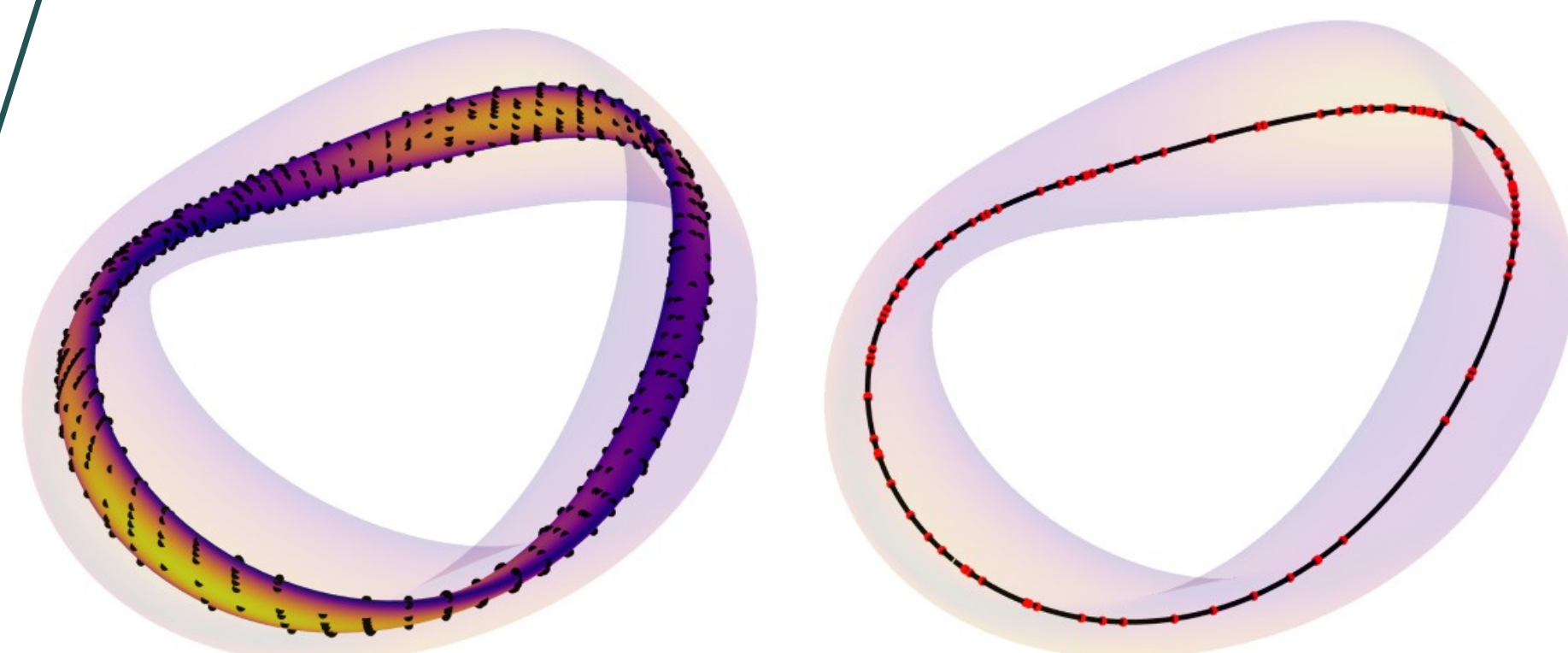
PARTICLE INITIALIZER

- Manual Initializers in lab/flux frame
- Initialize on a surface
- Initialize on a curve
- Initialize in volume based on fusion rate
- ...

PARTICLE TRACER

Single Function for all combinations

- Integration uses **Diffirax**¹
- Built upon **JAX**²
- Can run on CPU/GPU/TPU
- Automatically differentiable
- Many built-in solvers for **ODEs** and **SDEs**
- Different Adaptive Step-Size Controllers
- **Vectorized operations over particles**



¹ <https://github.com/patrick-kidger/diffrax>
² <https://github.com/jax-ml/jax>

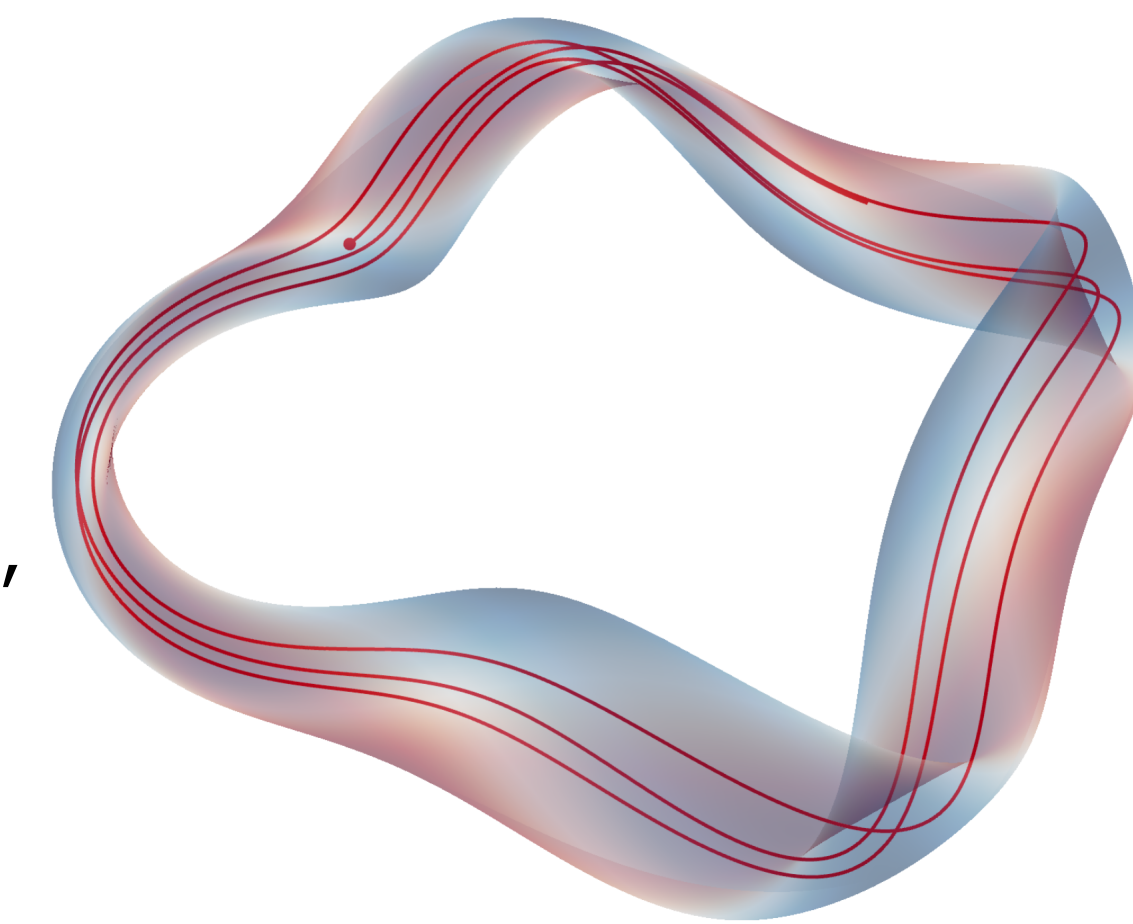
THEORY

The vacuum guiding center equations can be written as follows,

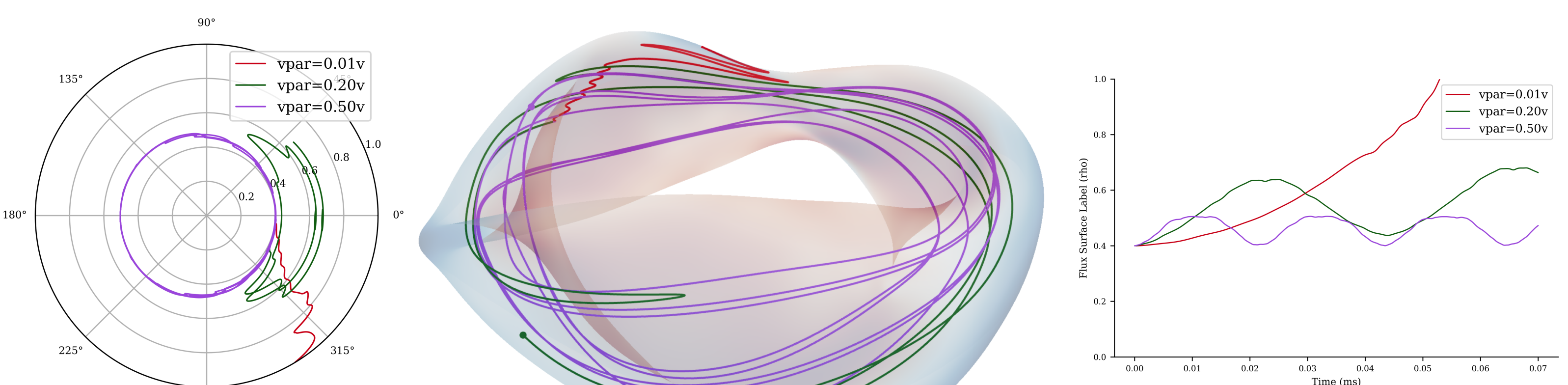
$$\frac{dR}{dt} = v_{\parallel} b + \frac{m}{qB^2} (v_{\parallel}^2 + 1/2 v_{\perp}^2) (b \times \nabla B)$$

$$\frac{dv_{\parallel}}{dt} = -\frac{v_{\perp}^2}{2B} (b \cdot \nabla B)$$

Single particle with $\rho_0=0.4$, $v_{par}/v=0.5$,
 $E=3.5 \times 10^6$ eV in scaled precise QH
equilibrium traced for 5e-5s



Particles can be traced in the equilibrium magnetic field without relying on interpolation, since DESC is fully spectral, we can compute all the terms necessary in the above equation without interpolation errors. For optimization, an interpolation based magnetic field will also be added as performant surrogate.

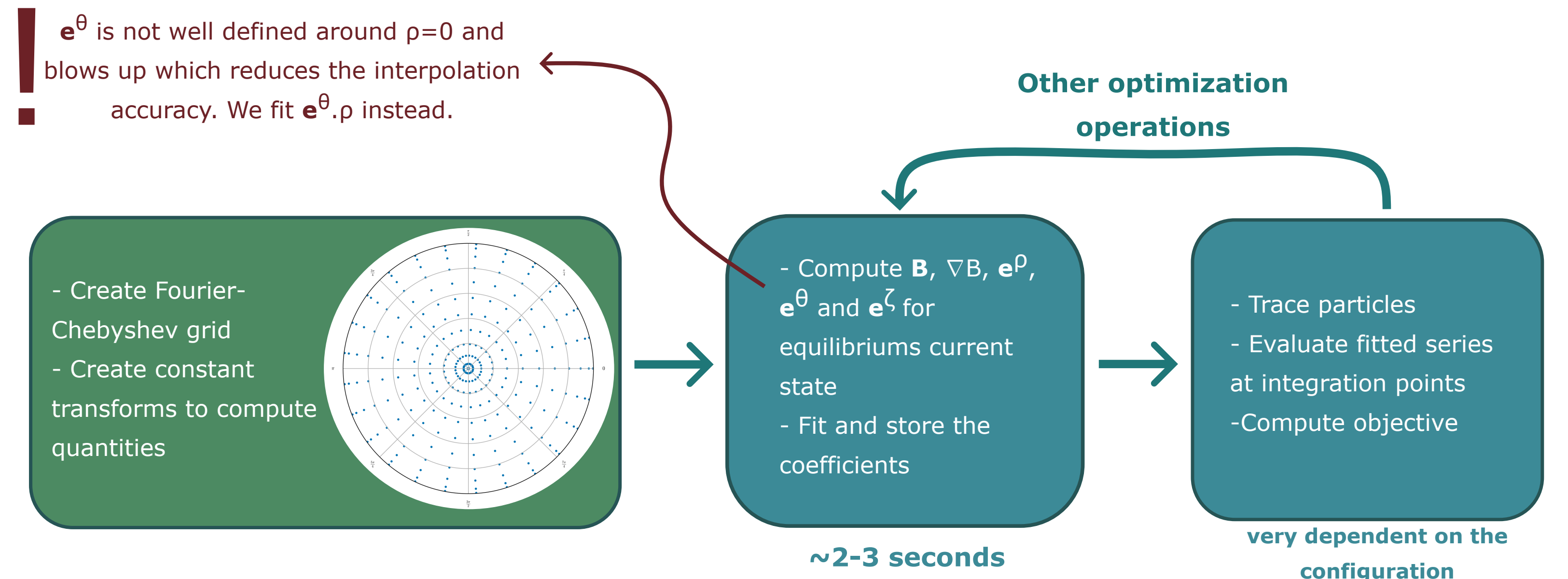


Three particles with $\rho_0=0.5$, $v_{par}/v=[0.01, 0.2, 0.5]$, $E=3.5 \times 10^6$ eV in scaled NCSX equilibrium traced for 7e-5s

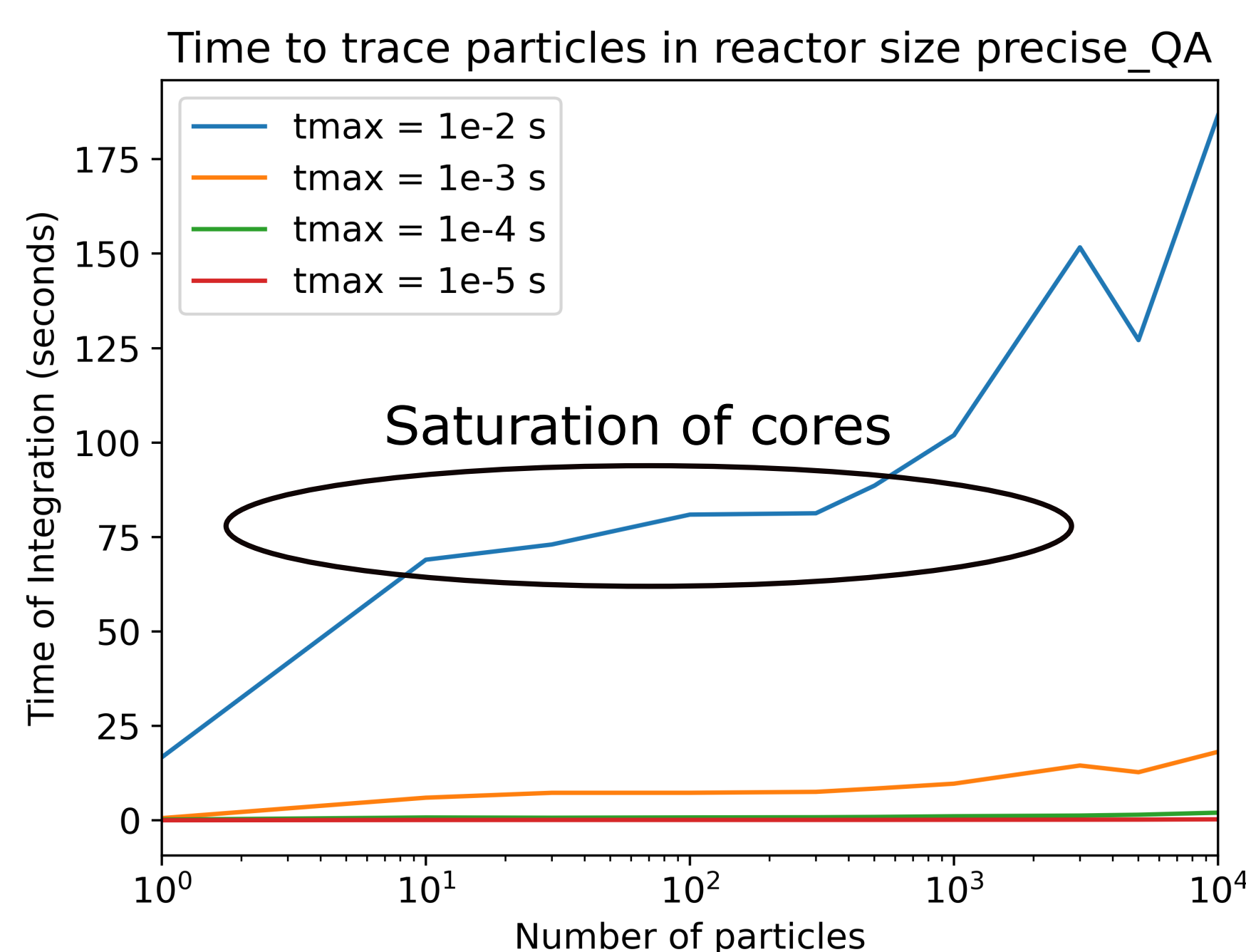
Fourier-Chebyshev Fit to the Field

$$\mathbf{B} = \frac{\partial \rho \psi}{2\pi \sqrt{g}} \left(\left(\iota - \frac{\partial \lambda}{\partial \zeta} \right) \mathbf{e}_{\theta} + \left(1 + \frac{\partial \lambda}{\partial \theta} \right) \mathbf{e}_{\zeta} \right) \quad B^i(\rho, \theta, \zeta) = \sum_l \sum_m \sum_n B_{lmn}^i \mathcal{F}_m(\theta) \mathcal{F}_n(\zeta) \mathcal{T}_l(\rho^*)$$

- Requires many Fourier-Zernike transforms
 - **Heavy in a particle tracing loop**
 - **Most accurate**
- **Faster in an optimization loop**
- Enough accuracy to get the trend for particles trajectory
- Requires 15 individual fits to compute guiding center equations



INTEGRATION TIME



Solver : 5th order explicit Runge-Kutta

Stepsize Controller : PID controller with 1e-4 absolute and relative tolerance

Minimum stepsize : 1e-8 seconds

Particles are created on $\rho=0.3$ surface randomly with random velocity ratio between -1 and 1.

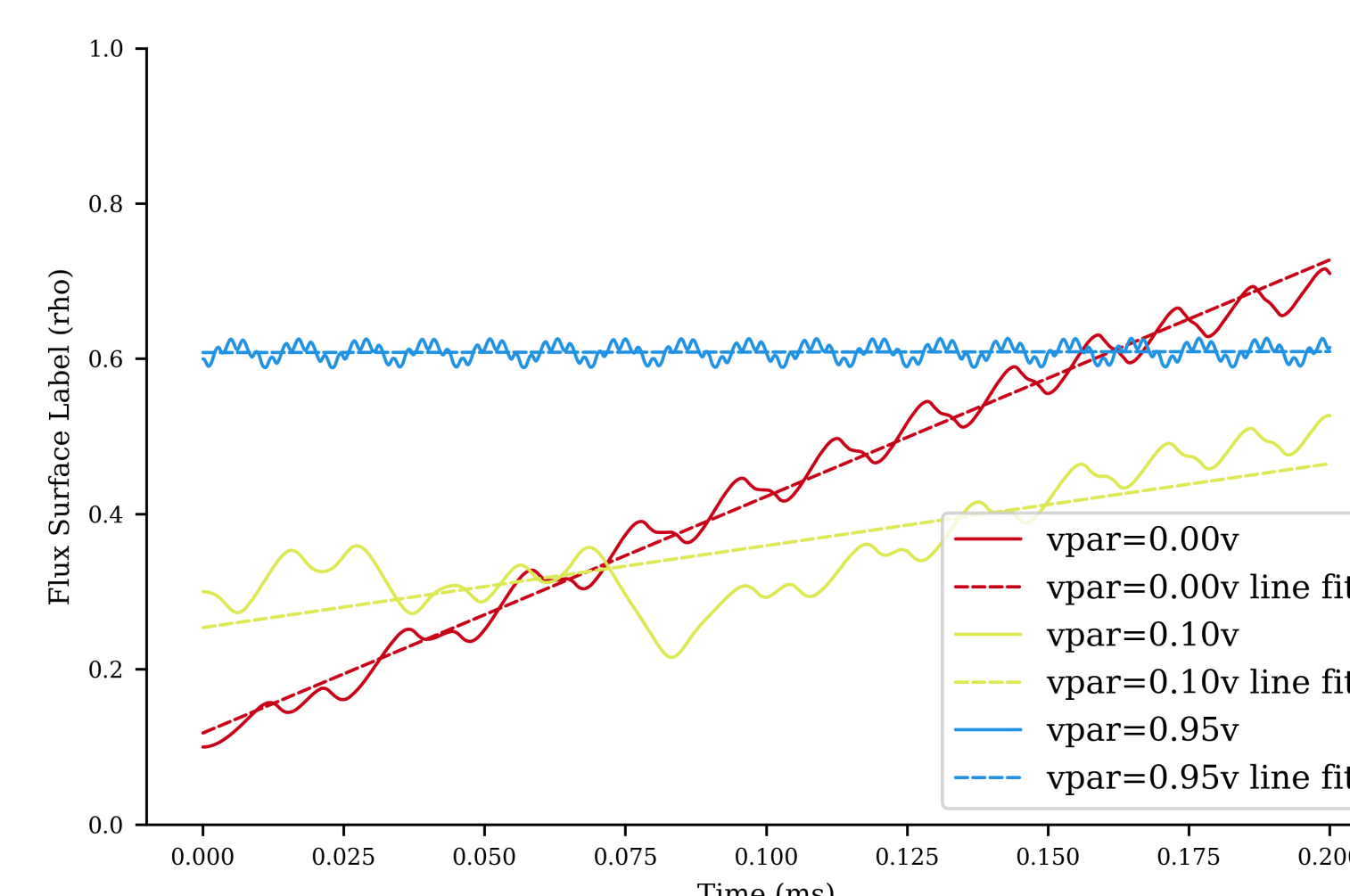
Particles are traced in the equilibriums magnetic field in DESC coordinates ρ, θ, ζ . No interpolation is done. Magnetic field and its gradient are computed at single point in a loop. **The integration time can be improved significantly by fitting a 3D Fourier-Chebyshev series or 3D splines to $\mathbf{B}$ and $\nabla \mathbf{B}$.**



PROPOSED OBJECTIVE

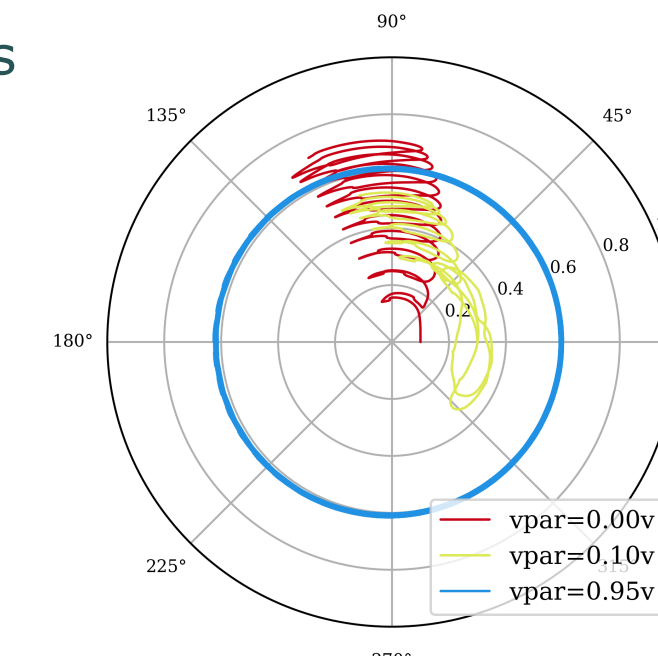
In literature, researchers used the prompt losses as the objective to minimize. In addition to that, we propose the new objective where the average slope of ρ vs time plot is used. This can reduce the need to integrate for longer times in exchange for more particles. As an example, below is the trajectories of 3 particles in scaled W7-X and the fitted lines.

$$at + b = [\rho - \rho_0] \rightarrow f(x) = dt_{total}$$



making sure that point (0,0) is in the set of points, helps to neutralize portion of the lost trajectories

remove the dependence on the integration time



An important computation is the gradient of the objective through **automatic differentiation** (AD). This is usually challenging since integration consists of for loops that are hard to apply chain rule for. **Diffirax** library is built for specifically this purpose. We support both forward and reverse mode AD.

